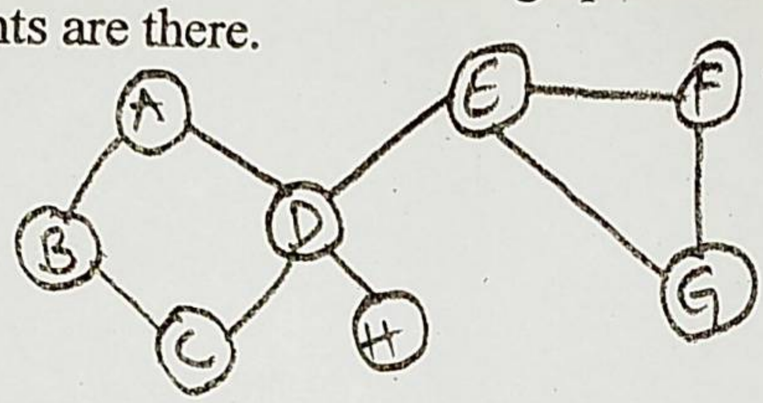
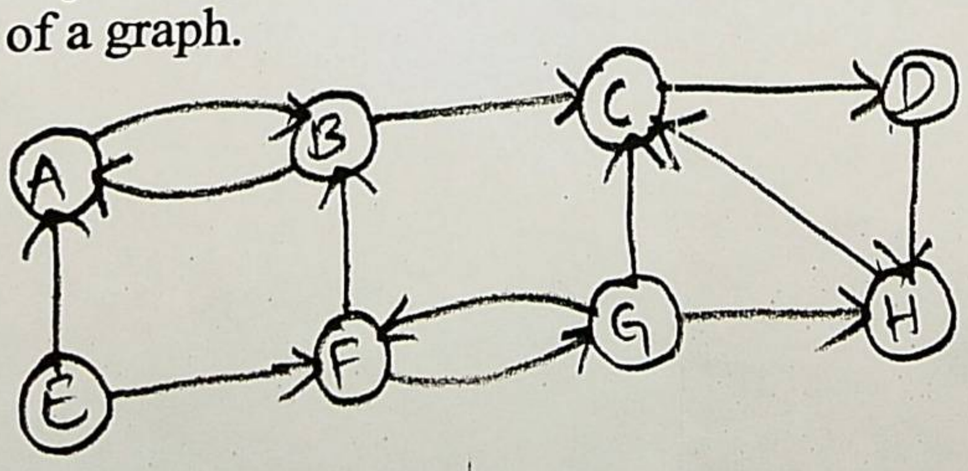
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
Osmania University, Hyderabad-07

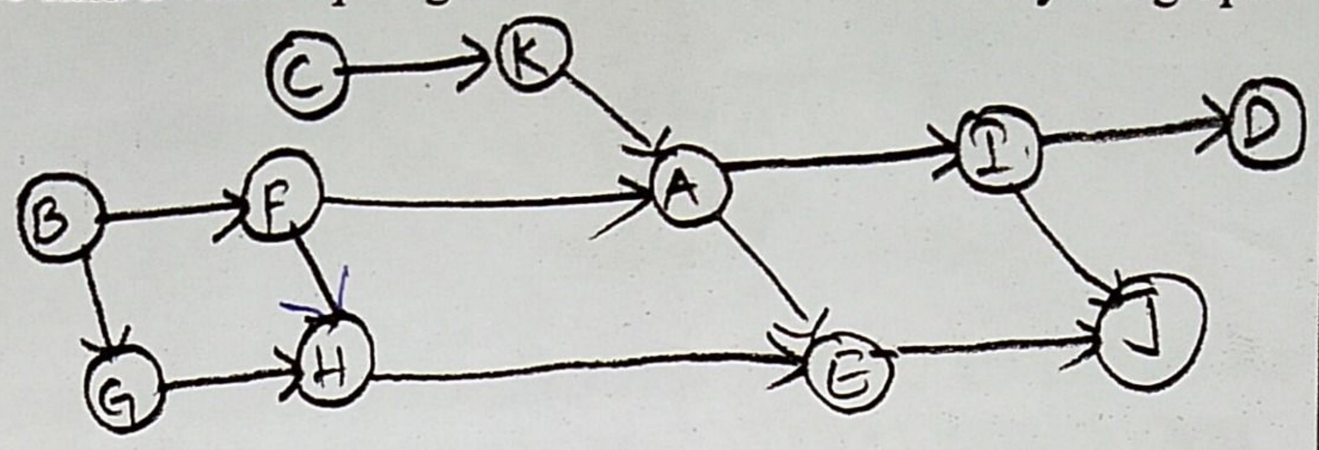
CLASS TEST-II

Subject: Design & Analysis of Algorithms

Class: B.E
Semester: IV
Academic Year: 2025-2026

Max Marks: 30
Duration: 1 1/2(9:30 a.m. to 11 :00 a.m.)
Date : 26-06-2025

	PART A Answer All Questions (5 x 2 = 10 M)	Marks	BT	CO
1. a)	Define articulation point. Given the below graph how many number of articulation points are there. 	2	2	2
b)	What are the different ways of representing graphs?	2	3	2
c)	Prove P⊆NP	2	3	3
d)	What is polynomial time reduction and what are its characteristics?	2	2	3
e)	Suppose you have a graph with 100 edges and 100 vertices. Is the graph sparse or dense, and should you represent this graph as an adjacency list or an adjacency matrix?	2	4	2
	PART B Answer Any Two Questions (2 x 10 = 20 M)			
2.	Discuss Kosaraju's algorithm to find the strongly connected components of a graph. 	10	2	2

3	How to find a valid topological ordering of a directed acyclic graph. 	10	2	
4	a) Show how to reduce an instance of the CNF-SAT problem to an equivalent instance of the 3SAT problem in polynomial time. b) Define the complexity class P, NP, NP hard, NP Complete	4+6	3	3

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